

HANUL HWANG

Stanford, CA $\diamond$ hanul@stanford.edu $\diamond$ (+1) 650 785 4109

ACADEMIC APPOINTMENTS

Physical Science Research Scientist

2025 – Present

Center for Turbulence Research (CTR), Stanford University, Stanford, CA

Postdoctoral Scholar

2022 – 2025

Center for Turbulence Research (CTR), Stanford University, Stanford, CA

EDUCATION

Ph.D. and M.S., Stanford University, Mechanical Engineering

2022

- Flow Physics and Computational Engineering (FPCE) Laboratory
- Dissertation: “Optimized amplification of interface distortion on liquid jets”
- Advisor: Professor Parviz Moin

B.S., Seoul National University, Mechanical & Aerospace Engineering

2016

- *Summa Cum Laude*
- Thesis Advisor: Professor Haecheon Choi

RESEARCH INTERESTS

Highly motivated in high-fidelity simulations of multiphysics, multiphase flows for engineering applications, including fuel atomization and heat transfer (cooling). Specific interests include:

- GPU-accelerated numerical algorithms for turbulent multiphase flows
- Computational physics modeling
- High-performance computing (HPC)

PEER-REVIEWED PUBLICATIONS

Journal Publications

1. **H. Hwang** and S. Park *et al.*, Highly dispersed nanomaterials in polymer matrix via aerosol-jet-based multi-material 3D printing, *Nano Energy*, 128, 109803, 2024.
2. **H. Hwang** and S. S. Jain, “A robust phase-field method for two-phase flows on unstructured grids”, *Journal of Computational Physics*, 507, 112972, 2024.
3. W. H. R. Chan, S. S. Jain, **H. Hwang** *et al.*, “Theory and simulations of linear and nonlinear two-dimensional Rayleigh–Taylor dynamics with variable acceleration”, *Physical Review Fluids*, 35, 042109, 2023.
4. **H. Hwang**, D. Kim and P. Moin, “Atomization of the optimally disturbed liquid jets”, *Physical Review Fluids*, 7 (11), 110508, 2022.
5. J. T. Kim, W. Ouyang, **H. Hwang**, H. Jeong *et al.*, “Dynamics of plosive consonants via imaging, computations, and soft-electronics”, *Proceedings of the National Academy of Sciences*, 119 (46), e2214164119.
6. **H. Hwang**, P. Moin and M. J. P. Hack, “A mechanism for the amplification of interface distortions on liquid jets”, *Journal of Computational Physics*, 911, 2021.
7. **H. Hwang**, W. H. R. Chan, S. S. Jain and S. I. Abarzhi, “Scale-dependent Rayleigh-Taylor dynamics with variable acceleration in a finite-sized domain for three-dimensional flows”, *Physical Review Fluids*, 33 (9) 092108, 2021.

Proceedings and Conference Papers

1. H. Song, **H. Hwang**, and P. Moin, “A Sharp-Interface Approach Based on Ghost Fluid Method for Compressible Two-Phase Flows”, AIAA SciTech Forum, 2026.
2. **H. Hwang** and C. Górlé, “Effects of curvature and interface advection calculation in two-phase wind and wave simulations” *Annual Research Briefs*, Center for Turbulence Research, Stanford University, 2024.
3. J. Wu and **H. Hwang**, “High-fidelity simulation of boundary layer flow over waves”, *Proceedings of the Summer Program*, Center for Turbulence Research, Stanford University, 2024.
4. W. H. R. Chan, A. Elnahas, **H. Hwang**, L. Brown and S. Balachandar, “Interscale analysis of sediment clusters amid turbulence”, *Proceedings of the Summer Program*, Center for Turbulence Research, Stanford University, 2024.
5. **H. Hwang** and C. Górlé, “Accurate wave generation using unstructured two-phase solver with a phase-field approach” *Annual Research Briefs*, Center for Turbulence Research, Stanford University, 2023.
6. **H. Hwang** and S. S. Jain, “A Phase-field method for simulations of two-phase flows on unstructured grids” *Annual Research Briefs*, Center for Turbulence Research, Stanford University, 2022.
7. S. S. Jain, E. Song, **H. Hwang** and P. Moin, “Continuum numerical modeling of rarefied flows with slip-wall boundary conditions” *Annual Research Briefs*, Center for Turbulence Research, Stanford University, 2020.
8. **H. Hwang**, W. H. R. Chan, S. S. Jain and S. I. Abarzhi, “Linear two-dimensional Rayleigh-Taylor dynamics with variable acceleration: comparison of theory and simulations”, *Annual Research Briefs*, Center for Turbulence Research, Stanford University, 2020.
9. **H. Hwang** and S. I. Abarzhi, “Scale-dependent Rayleigh-Taylor dynamics with variable acceleration in a finite-size domain for three-dimensional flows. Part 1”, *Annual Research Briefs*, Center for Turbulence Research, Stanford University, 2020.
10. **H. Hwang** and S. I. Abarzhi, “Scale-dependent Rayleigh-Taylor dynamics with variable acceleration in a finite-size domain for three-dimensional flows. Part 2”, *Annual Research Briefs*, Center for Turbulence Research, Stanford University, 2020.
11. **H. Hwang** and S. I. Abarzhi, “Scale-dependent Rayleigh-Taylor dynamics with variable acceleration in a finite-size domain for two-dimensional flows”, *Annual Research Briefs*, Center for Turbulence Research, Stanford University, 2020.
12. **H. Hwang** and M. J. P. Hack, “Optimization of the interface distortion of a round liquid jet”, *Annual Research Briefs*, Center for Turbulence Research, Stanford University, 107–120, 2019.
13. **H. Hwang** and M. J. P. Hack, “The most effective interface disturbance of a round liquid jet”, *Annual Research Briefs*, Center for Turbulence Research, Stanford University, 79–91, 2018.
14. **H. Hwang** and M. J. P. Hack, “Optimal disturbances in two-phase shear layers”, *Annual Research Briefs*, Center for Turbulence Research, Stanford University, 151–164, 2017.

MANUSCRIPTS/CONFERENCE IN PREPARATION

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1. **H. Hwang** and C. Górlé, “Assessing the Sensitivity of Coupled WindWave Simulations to Numerical Modeling Choices”, *36th Symposium on Naval Hydrodynamics*, 2026. *Accepted*
 2. S. R. Gomez, **H. Hwang**, and F. Zabaleta, “Predictions of turbulent windswell interactions using resolvent analysis”, *36th Symposium on Naval Hydrodynamics*, 2026. *Accepted*

3. **H. Hwang** and C. Górlé, “Manifestation of spurious currents and interface regularization in wind turbulence over fast-propagating waves”, *arXiv:2602.18024* *Under Review*
4. **H. Hwang** and C. Górlé, “Sharp interface method for windwave applications” *In Prep.*
5. M. Beeman, **H. Hwang** and C. Górlé “Wave loading on coastal structures” *In Prep.*
6. J. Wang, M. Beeman, **H. Hwang**, and C. Górlé “Wind injection location sensitivity in wind and wave flume” *In Prep.*

CONFERENCE PRESENTATIONS

1. **H. Hwang** and C. Górlé, “Revisiting the sharp-interface method for applications to windwave interactions”, *Annual Meetings of the Division of Fluid Dynamics of the American Physical Society*, Texas, USA, Nov. 2025.
2. H. Song, **H. Hwang**, and P. Moin, “A geometric-based sharp-interface approach for simulations of compressible multiphase flows”, *Annual Meetings of the Division of Fluid Dynamics of the American Physical Society*, Texas, USA, Nov. 2025.
3. J. Wang, **H. Hwang**, and C. Górlé, “Turbulence Development in Coupled Wind-Wave Flow: Wave Age Effects and Scaling Behavior”, *Annual Meetings of the Division of Fluid Dynamics of the American Physical Society*, Texas, USA, Nov. 2025.
4. M. Beeman, **H. Hwang** and C. Górlé, “Comparison of Two-Phase and Single-Phase Hurricane Wind Loading on Coastal Structures”, *Annual Meetings of the Division of Fluid Dynamics of the American Physical Society*, Texas, USA, Nov. 2025.
5. D. Rossinelli, Y. Hang, and **H. Hwang**, “CSF Flow in Complex Geometry: Validation of a High-Order CFD Transient Solver”, *Annual Meetings of the Division of Fluid Dynamics of the American Physical Society*, Texas, USA, Nov. 2025.
6. J. Wang, **H. Hwang** and C. Górlé, “Simulation of wind-wave interaction and its impact on wind loading on coastal buildings”, *15th Americas Conference on Wind Engineering*, Missouri, USA, May 2025.
7. **H. Hwang** and C. Górlé, “Confinement effects in high-speed wind and wave interactions”, *Annual Meetings of the Division of Fluid Dynamics of the American Physical Society*, Utah, USA, Nov. 2024.
8. W. H. R. Chan, A. Elnahhas, **H. Hwang**, L. Brown and S. Balachandar, “Interscale analysis of sediment clusters amid turbulence”, *Annual Meetings of the Division of Fluid Dynamics of the American Physical Society*, Utah, USA, Nov. 2024.
9. **H. Hwang** and C. Górlé, “Towards high-fidelity large-eddy simulations of extreme wind/wave events in coastal regions”, *16th International Conference on Wind Engineering*, Florence, Italy, 2023.
10. M. Beeman, **H. Hwang** and C. Górlé, “Validating multiphase numerical simulation of shoaling waves impacting a cylinder”, *16th International Conference on Wind Engineering*, Florence, Italy, 2023.
11. **H. Hwang** and C. Górlé, “Effects of turbulence in high-speed winds on wind-wave coupled simulations”, *Annual Meetings of the Division of Fluid Dynamics of the American Physical Society*, Indiana, USA, Nov. 2022.
12. M. Beeman, **H. Hwang** and C. Górlé, “Validation of CFD predictions of wave propagation and forces on a cylinder in a long flume”, *Annual Meetings of the Division of Fluid Dynamics of the American Physical Society*, Indiana, USA, Nov. 2022.

13. **H. Hwang**, D. Kim and P. Moin, “Atomization of optimally disturbed liquid jets”, *Annual Meetings of the Division of Fluid Dynamics of the American Physical Society*, Arizona, USA, Nov. 2021.
14. **H. Hwang**, P. Moin and M. J. P. Hack, “A new pathway for the amplification of distortions to the surface of liquid jets”, *Annual Meetings of the Division of Fluid Dynamics of the American Physical Society*, Chicago, Online, USA, Nov. 2020.
15. **H. Hwang**, P. Moin and M. J. P. Hack, “A new mechanism for the generation of interface distortions in liquid jets”, *Annual Meetings of the Division of Fluid Dynamics of the American Physical Society*, Washington, USA, Nov. 2019.
16. **H. Hwang** and M. J. P. Hack, “Optimal interface distortion of a round liquid jet”, *Annual Meetings of the Division of Fluid Dynamics of the American Physical Society*, Georgia, USA, Nov. 2018.

RESEARCH EXPERIENCE

Center for Turbulence Research, Stanford University

2025 – Present

Physical Science Research Scientist

- Develop and implement multi-physics models for multiphase flows, including flashing atomization, cavitation, and phase change, across incompressible and compressible regimes
- Design GPU-accelerated numerical algorithms (CUDA/HIP) for high-performance computing systems, with a focus on scalable parallelization and performance optimization
- Implement and evaluate advanced interface-capturing methods on unstructured grids, including volume-of-fluid (VOF), diffuse interface, level-set, edge-based interface tracking, and front-tracking techniques
- Investigate finite-thickness interface effects in comparison to sharp-interface approaches in multiphase, multi-physics simulations
- Formulate physics-based subgrid-scale models for interfacial flow dynamics

Center for Turbulence Research, Stanford University

2022 – 2025

Postdoctoral Scholar (PI: Prof. Parviz Moin, Prof. Catherine Gorlé)

- Large-eddy simulations for design of a National full-scale testing Infrastructure for Community Hardening in Extreme wind, surge, and wave events (NICHE) under National Science Foundation grant (CMMI-21311961)
- Investigated wind and wave interaction dynamics in extreme conditions
- Developed a robust and efficient multiphase solver

Joint research with Samsung Display Inc., Stanford University

2021 – 2022

Principal Investigator

- Investigated the capability of a slip-wall model in continuum flow limit for an accurate mass flux prediction for rarefied flows
- Developed continuum modeling of rarefied flows

Flow Physics & Computational Eng. Lab, Stanford University

2016 – 2022

MS & Ph.D (PI: Parviz Moin)

- Investigated the onset of the atomization of liquid jets and the effects of initial instability on the droplet distribution
- Optimization of interface distortions on liquid jets
- Conducted integrated simulations of atomization process of liquid jets

- Projects: Predictive Science Academic Alliance Program (PSSAP-III), Department of Energy and National Nuclear Security Administration (DE-NA0003968); Sub-Hinze scale breakup model for high-fidelity simulation of bubbly flows (#N00014-15-1-2726)

Turbulence, Flow Control and CFD/BioMimetic Eng. Lab, SNU

2014 – 2015

Research Intern (PI: Prof. Haecheon Choi)

- Investigated leapfrogging phenomenon regarding the initial relative positions of the two bubble rings and the velocity distribution
- Conducted two-phase flow simulations using level-set method as an interface tracking scheme coupled with TVD RungeKutta and WENO scheme

Micro Fluid Mechanics Lab, SNU

2015 – 2016

Research Intern (PI: Prof. HoYoung Kim)

- Built prototype search engine called “Zoogler” for applications of herd behavior of small insects to smallscale scout/monitoring robots

TEACHING, SERVICE & RECOGNITION

Teaching & Mentorship

- Guest Lecturer: Advanced Fluid Mechanics (ME 451B), Stanford Univ., Dec. 2022
 - Topic: Transient growth in non-orthogonal systems
- Teaching Associate: “Turbulence” with Prof. Parviz Moin (Spring 2018 & 2021)
 - Delivered 3+ hours/week lectures covering detailed derivations of class material
 - Provided hands-on experience with flow data analysis using canonical turbulent flows
 - Assisted in drafting the textbook “Fundamentals of Turbulent Flows” by Parviz Moin and Ronald Chan, Cambridge University Press
- Teaching Associate: “Advanced Topics in Turbulence” with Prof. Sanjiva K. Lele (Fall 2019)
 - Delivered weekly 1.5-hour lectures/discussions on advanced topics not covered in class

Professional Service

- Coordinator, Geophysical Fluid Mechanics subgroup, Summer Program, Center for Turbulence Research, Stanford University, July 2024
- Host, Summer Program, Center for Turbulence Research, Stanford University, July 2018 & 2022
- Stanford Thermal & Fluid Sciences Affiliates and Sponsors Program, Stanford University, Feb. 2018, 2019, 2020 and 2022

Honors & Fellowships

- American Physical Society, Division of Fluid Dynamics Gallery of Fluid Motion Award *2021*
 - Selected in a highly competitive process for a top-ranked video combining scientific merit and visual impact
- Franklin P. and Caroline M. Johnson Fellowship *2017–2018*
- Chunghap Scholarship *2010–2016*
 - Awarded for academic excellence; selected among top-ranked students
 - Full tuition support for B.S. studies at Seoul National University

CERTIFICATIONS

Argonne Training Program on Extreme-scale Computing

2024

- Competitively selected through an application process

- 80+ hours of lectures and hands-on sessions on HPC
- Topics included computer architectures, numerical algorithms, scalable community codes, and big-data methodologies

AI for Science on Supercomputers at Argonne Leadership Computing Facility *2022*

- Hands-on training in developing AI solutions on leadership-class supercomputers

OUTREACH AND SERVICE

Korean Students Association at Stanford(KSAS), Stanford Univ.

- President (2017–2018), Vice president (2017–2018), Financial officer (2019–2020)

Sergeant, Republic of Korea Army *2012 - 2014*

- Awarded for outstanding combat performance as a platoon leader trainee in the 459th Recruit Education Program at Nonsan Training Center
- Selected as the top squad leader at the 135th Squad Leader Training Center